

Forward and Reverse Automatic Differentiation by Hand

Learning goals

After this exercise, you should be able to

- explain what a dual number is and why dual numbers are useful for forward-mode automatic differentiation,
- propagate values and derivatives through a computation using dual numbers,
- compute directional derivatives and Jacobian columns with forward mode,
- explain the idea of a computational graph and an adjoint in reverse-mode automatic differentiation,
- propagate adjoints backwards through a computation graph,
- compute gradients and Jacobian rows with reverse mode,
- explain the practical difference between forward and reverse AD.

The calculations in this sheet are designed to be done by hand. No programming is required.

Part I - Forward-mode AD with dual numbers

1. What are dual numbers?

A dual number $\hat{x}$ has the form

$$\hat{x} = x + \dot{x}\varepsilon$$

where $x, \dot{x} \in \mathbb{R}$ and ε is a formal symbol with the special property

$$\varepsilon^2 = 0 \quad \varepsilon \neq 0$$

If we then evaluate a differentiable function f using the usual arithmetic rules together (think Taylor series) with $\varepsilon^2 = 0$, we obtain

$$f(x + \dot{x}\varepsilon) = f(x) + f'(x)\dot{x}\varepsilon$$

For the special choice $\dot{x} = 1$

$$f(x + \varepsilon) = f(x) + f'(x)\varepsilon$$

so the coefficient of ε is directly $f'(x)$.

2. Arithmetic with dual numbers

Let

$$\hat{a} = a + \dot{a}\varepsilon, \quad \hat{b} = b + \dot{b}\varepsilon$$

Then

Addition

$$\hat{a} + \hat{b} = (a + b) + (\dot{a} + \dot{b})\varepsilon$$

Multiplication

$$\hat{a}\hat{b} = (a + \dot{a}\varepsilon)(b + \dot{b}\varepsilon)$$

Expanding the product gives

$$ab + (a\dot{b} + \dot{a}b)\varepsilon$$

which is exactly the product rule.

Some useful elementary functions

$$\exp(a + \dot{a}\varepsilon) = e^a + \dot{a} e^a \varepsilon$$

$$\sin(a + \dot{a}\varepsilon) = \sin(a) + \dot{a} \cos(a)\varepsilon$$

$$\log(a + \dot{a}\varepsilon) = \log(a) + \frac{\dot{a}}{a}\varepsilon, \quad a > 0$$

and

$$(a + \dot{a}\varepsilon)^2 = a^2 + 2a \dot{a}\varepsilon$$

3. Very simple 1D example

Consider

$$f(x) = x^2 + 3x$$

and compute $f(2)$ and $f'(2)$ using dual numbers.

The dual $\hat{x}_0$ is (with tangent $\dot{x}_0 = 1$)

$$\hat{x}_0 = 2 + \varepsilon$$

Then

$$\hat{x}_0^2 = (2 + \varepsilon)^2 = 4 + 4\varepsilon$$

and

$$3\hat{x}_0 = 6 + 3\varepsilon$$

Therefore

$$f(\hat{x}_0) = 10 + 7\varepsilon$$

Hence

$$f(2) = 10$$

$$f'(2) = 7$$

The important point is that the value and the derivative were propagated at the same time.

4. More than one input

Suppose now that

$$g : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

We can seed every input with its own tangent component:

$$\hat{\mathbf{x}} = \mathbf{x} + \dot{\mathbf{x}}\varepsilon$$

Forward AD then gives

$$g(\mathbf{x} + \dot{\mathbf{x}}\varepsilon) = g(\mathbf{x}) + J_g(\mathbf{x}) \dot{\mathbf{x}} \varepsilon$$

where J_g is the Jacobian matrix.

Therefore, one forward-mode sweep computes one Jacobian-vector product

$$J_g(\mathbf{x})\dot{\mathbf{x}}$$

If we choose $\dot{\mathbf{x}}$ to be a standard basis vector, for example

$$\mathbf{e}_1 = (1, 0, 0)^T$$

then the tangent part is the first column of the Jacobian.

Forward-mode exercises

Exercise 1.1 - Scalar input, scalar output

Consider

$$f(x) = \log(xe^x + 3)$$

Evaluate the function and its derivative at

$$x = 1$$

using dual-number propagation.

Exercise 1.2 - Vector input, scalar output

Consider

$$g(x, y, z) = xy + \sin z + y^2$$

at the point

$$(x, y, z) = (1, 2, 0)$$

a.) First calculate the ordinary function value $g(1, 2, 0)$.

b.) Use forward-mode AD with the three seeds

$$\mathbf{e}_x = (1, 0, 0), \quad \mathbf{e}_y = (0, 1, 0), \quad \mathbf{e}_z = (0, 0, 1)$$

to determine the full gradient

$$\nabla g(1, 2, 0)$$

c.) Now use only one forward sweep with seed direction

$$\dot{\mathbf{x}} = (1, -1, 2)^T$$

to compute the directional derivative

$$D_{\dot{\mathbf{x}}}g = \nabla g^T \dot{\mathbf{x}}$$

Exercise 1.3 - Vector input, vector output

Consider the vector-valued function

$$\mathbf{h}(x, y, z) = \begin{pmatrix} h_1(x, y, z) \\ h_2(x, y, z) \end{pmatrix} = \begin{pmatrix} xy + z \\ x^2 + \sin y - z^2 \end{pmatrix}$$

Evaluate it at

$$(x, y, z) = (1, 0, 1)$$

a.) Calculate $\mathbf{h}(1, 0, 1)$.

b.) Use the three input basis directions

$$\mathbf{e}_x, \quad \mathbf{e}_y, \quad \mathbf{e}_z$$

and dual numbers to compute the full Jacobian

$$J_{\mathbf{h}} = \begin{pmatrix} \frac{\partial h_1}{\partial x} & \frac{\partial h_1}{\partial y} & \frac{\partial h_1}{\partial z} \\ \frac{\partial h_2}{\partial x} & \frac{\partial h_2}{\partial y} & \frac{\partial h_2}{\partial z} \end{pmatrix}$$

Remember: each forward sweep gives one column of the Jacobian.

c.) Use a single forward sweep with

$$\mathbf{v} = (1, 2, -1)^T$$

to compute

$$J_{\mathbf{h}} \dot{\mathbf{x}}$$

Part II - Reverse-mode AD / Backward AD

1. The basic idea

Forward mode propagates derivatives together with the values from input to output.

Reverse mode proceeds differently:

1. Perform an ordinary forward pass and store the intermediate values.
2. Start from the output.
3. Propagate sensitivities backwards through the computational graph.

For an intermediate variable v , define its adjoint as

$$\bar{v} = \frac{\partial L}{\partial v}$$

where L is the final scalar output whose derivative we want (the bar notation is common in reverse-mode AD).

The output is initialized with

$$\bar{L} = \frac{\partial L}{\partial L} = 1$$

Then the chain rule is applied locally, one operation at a time, in reverse order.

2. Local backward rules

Suppose an intermediate variable c is computed from earlier variables.

Addition

If

$$c = a + b,$$

then

$$\bar{a} += \bar{c}, \quad \bar{b} += \bar{c}$$

with $+=$ the programming operator meaning $\bar{a} + \bar{c}$ replaces the initial $\bar{a}$ -assignment: if one variable influences the output through several paths, all contributions to its adjoint must be added.

Multiplication

If

$$c = ab,$$

then

$$\bar{a} += \bar{c}b, \quad \bar{b} += \bar{c}a$$

Some useful elementary functions

$$c = e^a \quad \Rightarrow \quad \bar{a} += \bar{c} e^a$$

$$c = \sin a \quad \Rightarrow \quad \bar{a} += \bar{c} \cos a$$

$$c = \log a \quad \Rightarrow \quad \bar{a} += \bar{c} \frac{1}{a}$$

$$c = a^2 \quad \Rightarrow \quad \bar{a} += \bar{c} 2a$$

3. Very simple 1D reverse-mode example

Consider

$$f(x) = (x + 1)^2$$

at $x = 2$.

Break the function into elementary operations:

$$v_1 = x + 1$$

$$v_2 = v_1^2$$

$$f = v_2$$

Forward pass

At $x = 2$,

$$v_1 = 3, \quad v_2 = 9$$

Backward pass

Start with

$$\bar{v}_2 = 1$$

which is the condition $\frac{\partial f}{\partial v_2} \stackrel{!}{=} 1$.

Since

$$v_2 = v_1^2$$

we get

$$\bar{v}_1 = \bar{v}_2 2v_1 = 1 \cdot 6 = 6$$

and since

$$v_1 = x + 1$$

we obtain

$$\bar{x} = \bar{v}_1 = 6$$

Hence

$$f'(2) = 6$$

In one dimension this may look more complicated than ordinary differentiation. Its advantage becomes clear when a function has many inputs but only one scalar output.

4. Reverse mode for several inputs

For

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

one reverse sweep gives

$$\nabla f$$

with derivatives with respect to all inputs at once.

This is why reverse mode is particularly useful in machine learning: a neural network may have millions of parameters, but the loss is usually a single scalar.

For a vector-valued function

$$\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

we must first choose an output seed $\mathbf{w} \in \mathbb{R}^m$. Reverse mode then computes

$$J_{\mathbf{f}}^T \mathbf{w}$$

This is called a vector-Jacobian product or, equivalently under row-vector notation, $\mathbf{w}^T J$.

To reconstruct the complete Jacobian, use one reverse sweep per output basis vector.

Reverse-mode exercises

Use exactly the same three functions as in the forward-mode section.

Exercise 2.1 - Scalar input, scalar output

For

$$f(x) = \log(xe^x + 3)$$

at $x = 1$:

1. Write the function as a sequence of elementary intermediate variables.
2. Perform the forward pass and record all intermediate values.
3. Set the output adjoint equal to 1.

4. Propagate all adjoints backwards.

5. Determine $\bar{x} = f'(1)$.

Exercise 2.2 - Three inputs, scalar output

For

$$g(x, y, z) = xy + \sin z + y^2$$

at

$$(x, y, z) = (1, 2, 0)$$

use the intermediate variables

$$a = xy, \quad b = \sin z, \quad c = y^2$$

$$d = a + b + c$$

Exercise 2.3 - Three inputs, two outputs

For

$$\mathbf{h}(x, y, z) = \begin{pmatrix} xy + z \\ x^2 + \sin y - z^2 \end{pmatrix}$$

at

$$(x, y, z) = (1, 0, 1)$$

reverse mode needs an output seed because the output is not scalar.

a.) First output. Use

$$\mathbf{w}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

That is, propagate backwards only from h_1 .

Compute

$$J_{\mathbf{h}}^T \mathbf{w}_1$$

b.) Second output. Use

$$\mathbf{w}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Compute

$$J_{\mathbf{h}}^T \mathbf{w}_2$$

c.) Full Jacobian. Use the two results to reconstruct the complete Jacobian.

d.) General output seed. Without calculating the Jacobian from scratch, use

$$\mathbf{w} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

to calculate

$$J_{\mathbf{h}}^T \mathbf{w}$$

Short conceptual questions

Question 1

Why does setting $\varepsilon^2 = 0$ cause the coefficient of ε to behave like a derivative?

Question 2

For a function

$$f : \mathbb{R}^{100} \rightarrow \mathbb{R}$$

how many basis-direction forward sweeps would be needed to obtain the full gradient? How many reverse sweeps?

Question 3

For a function

$$f : \mathbb{R} \rightarrow \mathbb{R}^{100}$$

which mode would naturally be more efficient for constructing the full Jacobian?

Question 4

What is an adjoint $\bar{v}$?

Question 5

Why do reverse-mode updates use accumulation such as

$$\bar{x} += \dots$$

instead of simply assigning one value to $\bar{x}$?